

Chapter 4

Artifact: failure on a general probability space

The approximation property fails on general Ω

We now show that Proposition 4.3 of Bühler et al. does not extend verbatim to a general probability space. The example is a one-period market with a perfect (hence optimal) hedge whose hedge ratio is integrable but has no exponential moments.

Proposition 4.1. *There exist a one-period market on a general probability space, without transaction costs and without trading constraints, an entropic risk measure ρ_λ with arbitrary risk aversion $\lambda > 0$ and a position $X = -Z$ with $Z \in L^1$ such that*

$$\pi(X) \leq 0 \quad \text{and} \quad \pi_M(X) = +\infty \quad \text{for every } M \in \mathbb{N}.$$

Moreover, $\rho_\lambda(X + \delta_0(S_1 - S_0)) = +\infty$ for every strategy of the form $\delta_0 = F(I_0)$ with $F : \mathbb{R} \rightarrow \mathbb{R}$ bounded on $(0, 1)$; this covers all networks with bounded activation and all networks with continuous activation.

Proof. Let $\Omega := (0, 1) \times \{-1, +1\}$ with $\mathbb{P} := \text{Leb} \otimes (\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1})$, and write $U(\omega) := \omega_1$ and $\xi(\omega) := \omega_2$, so that U is uniform on $(0, 1)$, $\mathbb{P}[\xi = \pm 1] = \frac{1}{2}$ and U, ξ are independent. Take one trading period ($n = 1$), information $I_0 := U$, $I_1 := \xi$, hence $\mathcal{F}_0 = \sigma(U)$ and $\mathcal{F}_1 = \sigma(U, \xi)$, one asset ($d = 1$) with

$$S_0 := 1, \quad S_1 := 1 + \frac{\xi}{2},$$

no costs ($c_k \equiv 0$) and no constraints ($H_0(\delta_0) := \delta_0$). Note that S is a $\mathbb{P}$ -martingale and $S_1 > 0$. Define

$$h(u) := \frac{2}{\lambda} \log \frac{1}{u}, \quad u \in (0, 1), \quad Z := h(U) \frac{\xi}{2}, \quad X := -Z.$$

Then $\mathbb{E}[|Z|] = \frac{1}{2}\mathbb{E}[h(U)] = \frac{1}{\lambda} < \infty$, so $Z \in L^1$.

The perfect hedge. The strategy $\delta_0 := h(U)$ is $\mathcal{F}_0$ -measurable, and

$$X + \delta_0(S_1 - S_0) = -h(U)\frac{\xi}{2} + h(U)\frac{\xi}{2} = 0,$$

so $\pi(X) \leq \rho_\lambda(0) = 0$. Note that $\delta_0 = h(U) \in L^1$; the optimal strategy is integrable, so Hornik's theorem applies to it and networks approximate it in $L^1(\mathbb{P})$ and, along a subsequence, almost surely.

Every locally bounded strategy has infinite risk. Let $\delta_0 = F(U)$ with $m := \sup_{u \in (0,1)} |F(u)| < \infty$. Then

$$\rho_\lambda(X + F(U)(S_1 - S_0)) = \frac{1}{\lambda} \log \mathbb{E} \left[\exp \left(\frac{\lambda}{2} (h(U) - F(U)) \xi \right) \right].$$

The integrand is nonnegative, so the expectation is at least as large as its restriction to the event $\{\xi = +1\} \cap \{h(U) > m\}$. On that event one has $h(U) - F(U) \geq h(U) - m > 0$, hence

$$\exp \left(\frac{\lambda}{2} (h(U) - F(U)) \xi \right) = \exp \left(\frac{\lambda}{2} (h(U) - F(U)) \right) \geq \exp \left(\frac{\lambda}{2} (h(U) - m) \right) = \frac{e^{-\lambda m/2}}{U}.$$

Independence of U and ξ and $\mathbb{P}[\xi = +1] = \frac{1}{2}$ therefore yield, with $u_m := e^{-\lambda m/2}$,

$$\mathbb{E} \left[\exp \left(\frac{\lambda}{2} (h(U) - F(U)) \xi \right) \right] \geq \frac{e^{-\lambda m/2}}{2} \int_0^{u_m} \frac{1}{u} du = \infty,$$

so the entropic risk is $+\infty$.

Networks are locally bounded. If the activation σ is bounded, every network $F = W_L \circ F_{L-1} \circ \dots \circ F_1$ is bounded on all of $\mathbb{R}$, because the output of the last hidden layer lies in a bounded set and W_L is affine. If σ is continuous, F is continuous on $\mathbb{R}$ and therefore bounded on the compact set $[0, 1] \supseteq (0, 1)$. In either case, every network strategy $\delta_0 = F(I_0)$ has $\sup_{(0,1)} |F| < \infty$, so $\rho_\lambda(X + W(\delta)) = +\infty$ and $\pi_M(X) = +\infty$ for every M . $\square$